20 Magnetic materials and their properties

20.1 Magnetic materials

All materials have magnetic properties. These characteristic properties may be divided into five groups: diamagnetic, paramagnetic, ferromagnetic, antiferromagnetic and ferrimagnetic. Only the ferromagnetic and ferrimagnetic materials have properties which are useful in practical applications.

Ferromagnetic properties are confined almost entirely to iron, nickel and cobalt in their alloys. The only exceptions are some alloys of manganese and some of the rare earth elements. Of these gadolinium has the highest Curie temperature, about 16°C (see the Appendix for definitions and conversion factors). Some of the magnetic properties of the ferromagnetic elements including the rare earths are given in Tables 20.1 and 20.2. Ferrimagnetism is the magnetism of the oxides of the ferromagnetic elements. These are variously called ferrites and garnets. The basic ferrite is magnetite (Fe_3O_4) which can be written as $FeO \cdot Fe_2O_3$. By substituting for the FeO with other divalent oxides a wide range of compounds with useful properties can be produced. The main advantage of these materials is that they have high electrical resistivity which minimises eddy currents when they are used at high frequencies.

Ferromagnetic and ferrimagnetic materials are characterised by moderate to high permeabilities (*see* Tables 20.4, 20.5 and 20.7–20.11). The permeability varies with the applied magnetic field, rising to a maximum at the knee of the *B–H* curve and reducing to a low value at very high fields. They also exhibit magnetic hysteresis whereby the intensity of magnetisation of the material varies according to whether the applied field is being increased in a positive sense or decreased in a negative sense. A typical hysteresis loop is shown in Figure 20.1 together with some of the more important derived magnetic properties. When the magnetisation is cycled continuously round a hysteresis

Table 20.1 FERROMAGNETIC ELEMENTS[9]

	$J_s(T)$		*Curie temperature*		
	20°C	4K	°C	σ_{4K}	$\sigma_{20°C}$
Fe	2.153	2.193	770	221.7	217.6
Ni	0.617	0.656	358	58.6	55.1
Co	1.790	1.797	1 121	162.5	161.9
Gd	0	2.47	16	250	0

Table 20.2 RARE EARTH ELEMENTS[9]

	Gd	Tb	Dy	Ho	Er	Tm
$J_s(T)$ 4 K	2.47	3.43	3.75	3.81	3.41	2.70
σ_0 4 K	250	330	350	345	300	230
T_c K	289	222	85	20	20	25

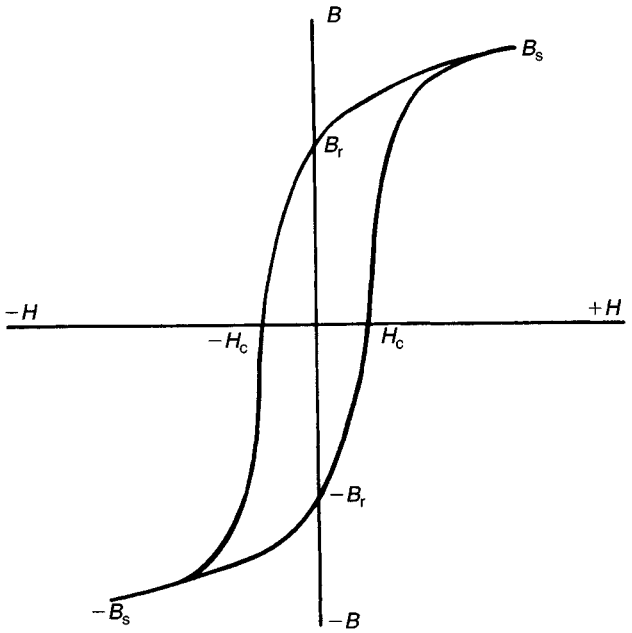


Figure 20.1 *Typical hysteresis loop*

loop, as for example when the applied field arises from an alternating current, there is an energy loss proportional to the area of the included loop. This is termed hysteresis loss, and is measured in joules per metre.[3] High hysteresis loss is associated with permanent magnetic properties exhibited by materials commonly termed 'hard' magnetic materials as these often have hard mechanical properties. Those materials with low hysteresis loss are termed 'soft' and are difficult to magnetise permanently. The ferromagnetic properties of these elements, alloys, and compounds disappear reversibly if heated above the Curie temperature, when they become paramagnetic.

Paramagnetic substances of elongated shape tend to align themselves with an applied magnetic field as do the ferromagnetics, but do not exhibit any permanent magnetic properties when the field is removed as hysteresis is absent. Their magnetic susceptibility is very small but positive, and consequently their relative permeability is only slightly greater than unity. The susceptibility is generally independent of field strength but may decrease with increasing temperature. Among the paramagnetic substances are included many iron salts and rare-earth elements salts, palladium and platinum, sodium, potassium, and oxygen, and the ferromagnetics above their Curie point.

The diamagnetic substances take on a very small magnetisation which is opposed to the sense of the applied field. Thus an elongated specimen will align itself transversely to a non-uniform field. The diamagnetics have negative susceptibilities and consequently permeabilities which are very slightly less than unity. Many of the metals and most non-metals are diamagnetic. It is usual to compare paramagnetic and diamagnetic substances by their susceptibilities rather than by permeability or saturation magnetisation as for the ferromagnetic materials. The mass susceptibilities of the elements at room temperature are given in Table 20.3.[5–9]

For bulk applications of magnetic material, iron, steel or cast iron are used. They have the advantage of cheapness and strength. These materials should only be used for d.c. applications as they have low electrical resistivity which would give rise to eddy currents if used in alternating fields. The properties of some steels and cast irons are given in Tables 20.4 and 20.5.[10–13]

For a list of current reviews, at the time of writing, on magnetic materials see references 14–19.

20.2 Permanent magnet materials[20,21]

The properties of a permanent magnet material are given by the demagnetisation curve, Figure 20.2, the second quadrant of the *B–H* hysteresis loop. This extends from the remanence B_r to the coercivity

Table 20.3 MASS SUSCEPTIBILITIES OF THE ELEMENTS[5–9]

Element	*Atomic No.*	*Mass susceptibility* 10^{-8} m^3 kg^{-1}	*Element*	*Atomic No.*	*Mass susceptibility* 10^{-8} m^3 kg^{-1}
H	1	−2.48	Ag	47	−0.23
He	2	−0.59	Cd	48	−0.23
			In	49	−0.14
Li	3	+0.63	Sn	50	−0.31
Be	4	−1.26	Sb	51	−1.09
B	5	−0.87	Te	52	−0.39
C	6	−0.62	I	53	−0.45
N	7	−1.00	Xe	54	−0.43
O	8	+133.5			
F	9		Cs	55	−0.28
Ne	10	−0.41	Ba	56	+1.13
			La	57	+1.02
Na	11	+0.64	Ce	58	+22.0
Mg	12	+0.69	Pr	59	+42.3
Al	13	+0.82	Nd	60	+48.0
Si	14	−0.16	Pm	61	—
P	15	−1.13	Sm	62	+11.1
S	16	−0.62	Eu	63	+27.6
Cl	17	−0.72?	Gd	64	*
Ar	18	−0.60	Tb	65	+1360
			Dy	66	+545
K	19	+0.65	Ho	67	+549
Ca	20	+1.38	Er	68	+377
Sc	21	+8.80	Tm	69	+199
Ti	22	+4.21	Yb	70	+0.59
V	23	+6.28			
Cr	24	+4.45	Lu	71	+0.12
Mn	25	+12.2	Hf	72	+0.53
Fe	26	*	Ta	73	+1.07
Co	27	*	W	74	+0.39
Ni	28	*	Re	75	+0.46
			Os	76	+0.06
Cu	29	−0.10	Ir	77	+0.23
Zn	30	−0.20	Pt	78	+1.22
Ga	31	−0.30			
Ge	32	−0.15	Au	79	−0.18
As	33	−0.39	Hg	80	−0.21
Se	34	−0.40	Tl	81	−0.30
Br	35	−0.49	Pb	82	−0.15
Kr	36	−0.44	Bi	83	−1.70
			Po	84	—
Rb	37	+0.26	At	85	—
Sr	38	−0.25	Rn	86	—
Y	39	+6.66			
Zr	40	+1.66	Fr	87	—
Nb	41	+2.81	Ra	88	—
			Ac	89	—
Mo	42	+1.17	Th	90	+0.53
Tc	43	+3.42	Pa	91	+3.25
Ru	44	+0.54	U	92	+2.15
Rh	45	+1.32	Np	93	—
Pd	46	+6.57	Pu	94	+3.14
			Am	95	+5.15

H_{cB}. It can be shown that when a piece of permanent magnet material is put into a magnetic circuit, the magnetic field generated in a gap in the circuit is proportional to BHV, where B and H are the corresponding points at a point on the demagnetisation curve and V is the volume of permanent magnet. So to obtain a given field with the minimum volume of magnet material we require the product BH to be a maximum. The magnet is then designed so that its B, H value is as close as possible to the $(BH)_{max}$ value. It is also useful to use the BH_{max} value to compare the characteristics

Table 20.4 MAGNETIC PROPERTIES OF WROUGHT STEELS[12–13]

*Grade**	*En No*	*Material*	*Flux density* T		*Rel. permeability*		*Max. flux density* T	*Remanence* T	*Coercive force* A m^{-1}
			H = 4 kA m^{-1}	H = 8 kA m^{-1}	Maximum at	*H* A m^{-1}	H = 20 kA m^{-1}		
220M07	1A	Free cutting carbon steel C 0.15 max., Si 0.40 max., Mn 0.90–1.30, S 0.20–0.35	1.50	1.76	1 200	800	1.95	0.87	240–400
040A04	2A/1	Low carbon steel C 0.08 max., Si 0.40 max., Mn 0.50 max., S and P 0.05 max.	1.67	1.77	2 400–5 000	400–160	1.96	0.5–1.1	40–120
070M20	3	Carbon steel C 0.25 max., Si 0.05–0.35, Mn 1.00 max., S and P 0.05 max.	1.57	1.74	1 100	800	1.96	1.00	240–400
080M30	5	'30' carbon C 0.25–0.35, Si 0.05–0.35, Mn 0.60–1.00, S and P 0.05 max.	1.52	1.73	1 000	800	1.88	—	800
070M55	9	'55' carbon C 0.50–0.60, Si 0.10–0.40, Mn 0.50–0.90, S and P 0.05 max.	1.45	1.67	400	2 000	1.81	1.30	1 200
Low alloy steels		C 0.20–0.60, Si 0.1–0.40, Mn 2.0 max., S and P 0.05 max., Ni 1.0 max., Cr 1.0 max.	1.40–1.65	1.70–1.90	400–650	—	1.75–1.90	0.88–1.55	600–1 600
3% Nickel steel		C 0.20–0.60, Si 0.10–0.40, Mn 0.8 max., S and P 0.05 max., Ni 2.5–3.0, Cr 1.0 max., Mo 0.5 max.	1.50–1.70	1.73–1.85	600	1 600	1.85–1.95	1.43	980
3% Chromium steel		C 0.20–0.60, Si 0.10–0.40, Mn 0.8 max., S and P 0.05 max., Cr 3.0, Ni 0.40 max.	1.65	1.75	680	1 600	1.75–1.85	1.50	880
Chromium rust resisting steels		C 0.35 max., Si 0.1 max., Mn 1.0 max., S and P 0.05 max., Ni 3.0 max., Cr 12–20	0.95–1.30	1.06–1.40	660	720	1.43–1.64	0.60–1.00	400–1 250

Information from 'The Mechanical and Physical Properties of British Standard En Steels' by J. Woolman and R. A. Mottram, volumes 1, 2 and 3, Pergamon, Oxford, 1964.[12]
*The numbered grades refer to BS 970:PART 1:1983 (*see* Ref. 13).

Table 20.5 SOME DATA ON THE MAGNETIC PROPERTIES OF CAST STEELS AND CAST IRONS

Material	*Condition*	*Flux density* T		*Rel. permeability*		*Max. flux density* T	*Remanence* T	*Coercive force* $A\,m^{-1}$
		$H = 4\,kA\,m^{-1}$	$H = 8\,kA\,m^{-1}$	Maximum at H $A\,m^{-1}$		$H = 20\,kA\,m^{-1}$	$H = 20\,kA\,m^{-1}$	$H = 20\,kA\,m^{-1}$
Cast steel[11]								
0.10% carbon steel,	Annealed	1.62	1.74	2 420	216	1.92	0.80	136
Si, 0.33%; Mn, 0.67%	Normalised	1.64	1.76	1 950	248	1.96	0.85	168
	As-cast	1.61	1.74	2 100	208	1.95	0.66	128
0.19% carbon steel,	Annealed	1.60	1.73	2 100	224	188	0.87	156
Si, 0.30%; Mn, 0.48%	Normalised	1.57	1.72	1 520	360	1.91	0.90	216
	As-cast	1.55	1.69	1 720	232	1.89	0.73	168
0.34% carbon steel,	Annealed	1.53	1.66	1 200	400	1.85	1.05	296
Si, 0.44%, Mn, 0.55%	Normalised	1.50	1.67	970	575	187	1.05	440
	As-cast	1.45	1.65	840	480	1.87	0.85	440
Manganese steel								
0.19% C, 0.48% Si, 1.14% Mn	A 925°C	1.51	1.66	1 300	400	1.86	—	—
0.29% C, 0.29% Si, 1.40% Mn	A 950°C, N 880°C, T 600°C, a.c.	1.50	1.68	650	960	1.78	—	—
0.31% C, 0.52% Si, 1.37% Mn	A 950°C, OQ 860°C, T°629°C, a.c.	1.55	1.65	750	960	1.70	—	—
Chromium-molybdenum steel								
0.31% C, 0.69% Mn	A 925°C, N 880°C, T 700°C, a.c.	1.45	1.59	—	—	1.76	—	—
1.16% Cr, 0.39% Mo	A 925°C, OQ 860°C, T 650°C, a.c.	1.54	1.66	—	—	1.81	—	—
						Hysteresis loss JM^{-3} $cycle^{-1}$		
Cast-irons[10]								
Grey iron T.C. 3.12%, Si 2.2%,	As-cast	0.70	0.86	315	640	2 700	0.41	560
Mn 0.67%, P 0.13%	Annealed	0.93	1.07	1 560	200	700	0.44	200
(low phosphorus)								
Grey iron T.C. 3.3%, Si 2.04%,	As-cast	0.79	0.97	281	1 040	2 730	0.43	720
Mn 0.52%, P 1.03%	Annealed	0.85	1.00	760	320	1 190	0.44	280
(high phosphorus)								
Whiteheart Malleable								
TC 1.01%, Si 0.66%, Mn 0.25%	Pearlitic center	1.32	1.42	730	560	1 490	0.75	360
TC 0.46%, Si 0.66%, Mn 0.25%	Mainly ferritic	1.35	1.61	1 455	280	840	0.74	200
Blackheart Malleable		1.40	—	2 120	176	490	0.62	120
TC 1.26%, Si 0.83%, Mn 0.26%								
Spheroidal Graphite	Pearlitic	0.93	1.12	290	1 200	3 040	0.54	920
TC 2.9%, Si 2.61%, Mn 0.72%, Ni 2.18%	Annealed, ferritic	1.14	1.26	1 150	320	700	0.46	200
Spheroidal Graphite TC 3.64%,	Annealed	1.23	1.42	2 060	240	450	0.61	120
Si 1.41%, Mn 0.3% (nickel free)								

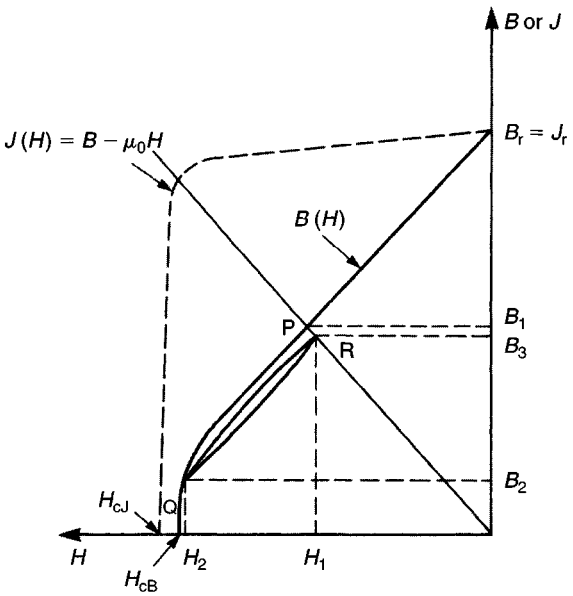


Figure 20.2 *Demagnetisation curve and recoil loop*

of materials. Generally the material with the highest $(BH)_{max}$ will be chosen but this has to be weighed against such considerations as cost, shape, manufacturing problems and stability.

When the magnet is fully magnetised in a completed magnetic circuit, the working point is the remanence B_r. When a gap is made in the circuit, the magnet will be partly demagnetised and its working point will fall to, say, the point P. If now a further demagnetising field is applied to the magnet corresponding to H_1–H_2 the flux density will be reduced to B_2. If this extra field H_1–H_2 is removed the field will recoil along the narrow loop QR to the point R. The corresponding flux density is B_3. This narrow loop is called the recoil loop and its slope is $\mu = \mu_0\mu_r$ where μ_r is the relative recoil permeability and μ_0 is the magnetic constant $(= 4\pi \times 10^{-7})$. This is an important consideration in dynamic applications such as motors, generators and lifting devices where the working point of the magnet changes as the magnetic circuit configuration changes. It is also important when considering premagnetising before assembly into a magnetic circuit. If μ_r is nearly equal to unity as it is for ferrites and for rare earth alloys then the magnets can be premagnetised before assembly into a magnetic circuit without much loss of available flux. However, if μ_r is considerably greater than unity, care must be taken to magnetise the magnet in the assembled magnetic circuit.

The dotted curve in Figure 20.2 is the J–H curve where $B = \mu_0 H + J$. For materials with B_r much greater than $\mu_0 H_{cB}$, e.g. Alnico, the two curves are almost identical, but for ferrites and rare earth alloys where B_r and $\mu_0 H_{cB}$ are close in value the curves become quite different. The J–H curve has only a very gradual slope from B_r; this indicates that the magnetisation in the material is nearly uniform and a new parameters H_{cJ} is introduced. This intrinsic coercivity H_{cJ} is a measure of the difficulty or ease of demagnetisation. As the value of H_{cJ} is increased the more the material will resist demagnetisation due to stray fields, etc. However, as a field of at least three times H_{cJ} must be applied to the magnet for magnetisation, difficulties may be encountered in attaining the full magnetisation if H_{cJ} is too high.

Magnets in the three main groups Alnico, ferrites and rare earth alloys are generally made anisotropic with the properties in one direction considerably better than in the other directions. This best direction is called the preferred direction of magnetisation. The curves and parameters supplied for such a material refer to the properties in this preferred direction.

To choose between these three materials for a particular application, Alnico must be used where magnetic stability is a requirement and this would apply in any type of instrument application. Ferrite will be used where cost is the main consideration and rare earth magnets will be used if the highest possible strength is required or if miniaturisation is needed.

20.2.1 Alnico alloys

A wide range of alloys with magnetically useful properties is based on the Al–Ni–Co–Fe system. They are characterised by high remanence, available energy and moderately high coercivity. They are very stable against vibration and have the widest useful temperature range (up to over 500°C) of any permanent magnet material. But they are mechanically hard, impossible to forge, and difficult to machine except by grinding and special methods such as spark erosion.

The preferred form of magnet is a relatively simple shape made by casting or sintering, precision ground on essential surfaces only. Soft-iron pole pieces may be clamped on or held by a suitable adhesive such as Araldite. Soldering, diecasting and even brazing (with precautions against overheating) may be used.

The commonest alloys contain 23–25% cobalt, 12–14% nickel, about 8% aluminium, a few percent copper and sometimes small additions of niobium and silicon, with the balance iron. They are cooled at a controlled rate in a magnetic field applied in the direction in which the magnets are to be magnetised. The properties are much improved in this direction at the expense of those in other directions. There is finally a fairly prolonged and sometimes rather complicated treatment at temperatures in the range 650–550°C.

The field cooling is much more effective in producing anisotropic properties if the magnets are cast with a columnar structure. Basically a columnar structure is produced by casting the alloy into a hot mould with a cold chill. The mould may be made of special materials which can be preheated to at least 1 250°C in a furnace, or alternatively contain chemicals which heat it by an exothermic reaction. Either method is inconvenient and expensive; there are also limitations on the lengths and shapes of magnets that can be made columnar. This Columax is therefore regarded as a rather expensive, quality material.

The coercivity of Alnico can be improved by a factor of two to three by increasing the cobalt content to 30–40% and adding 5–8% of titanium with possibly some niobium. For the best properties it is necessary to hold these alloys for several minutes in a magnetic field at a constant temperature, accurately controlled to ±10°C, instead of the usual field cooling process. This heat treatment as well as the high cobalt content makes these alloys expensive and they are only used where the higher coercivity is required. Columnar versions of these alloys can also be made but these are only used for very specialised applications.

The Alnico alloys can be produced as castings of up to 100 kg and down to a few grams in weight but it is found more economical to produce the smaller sizes of 50 g and less by the sintering process. In all cases it is advisable to contact the manufacturers before any design is considered.

20.2.2 Ferrite

The permanent magnet ferrites (also called ceramics) are mixed oxides of iron (ferric) oxide with a divalent heavy metal oxide, usually either barium or strontium. These ferrites have a hexagonal crystal structure, the very high anistropy of which gives rise to high values of coercivity, e.g. 150–300 kA m^{-1} (compared with about 110 kA m^{-1} for the best Alnico alloys). The general formula is $MO \cdot 5.9Fe_2O_3$ where M is either barium or strontium, and the crystal structure is called magneto plumbite as it was originally found in the equivalent lead oxide compound. These ferrites are made by mixing together barium or strontium carbonate with iron oxide in the correct proportions. The mixture is fired in a mildly oxidising atmosphere and the resulting mixture is milled to a particle size of about 1 micron. This powder is then pressed in a die to the required shape (with a shrinkage allowance), and anisotropic magnets are produced by applying a magnetic field in the direction of pressing. After pressing, the compact is fired. This material is a ceramic and can only be cut by high speed slitting wheels. Ferrite magnets are produced in large quantities in a variety of sizes for different applications. Flat rings are made for loudspeakers ranging up to 300 mm in diameter with a thickness of up to 25 mm. Segments are made for motors with ruling diameters from 40 to 160 mm and rectangular blocks are made for separators with dimensions of up to 150 × 100 × 25 mm. These blocks can be built up into assemblies or cut down into smaller pieces for a variety of applications. In each of these cases the preferred direction of magnetisation is through the shortest dimension. Generally the magnetic length does not exceed 25 mm and if a longer length is required this is built up with magnets in series.

The great success of permanent magnet ferrites is due to the low price per unit of available magnetic energy, the high coercivities, the high restivities and the low density. The isotropic grades are the least expensive to manufacture and may be magnetised into complex pole configurations; they are used for a wide range of relatively small-scale applications. The largest application in terms of market volume is for magnets for loudspeakers; here the high-remanence anisotropic grades

are used. The other high-volume application is for field magnets in small d.c. motors, particularly those used in the automotive industry, e.g. for windscreen wipers, blowers, etc. In this application the high-coercivity anisotropic ferrite is formed into an arc-shaped segment, magnetised radially; a pair of segments embrace the armature and provide the stator field. Other applications of anisotropic hard ferrites include magnetic chucks and magnetic filters and separators.

20.2.3 Rare earth cobalt alloys

These are the results of a fairly recent development and the first commercial material was $SmCo_5$. The alloy powder is produced either by reducing the rare earth powder together with cobalt powder or by preparing the alloy and powdering it down. The powder is compacted and pressed, almost invariably in a magnetic field, to produce an anisotropic compact. This is then sintered at 1 155±15°C and followed by a tempering treatment at about 900°C. It must be emphasised that during all the processing of these rare earth alloys a protective atmosphere is used to prevent the oxidation of the rare earth metals which are very reactive.

Another development in this series is the Sm_2Co_{17} composition which is really the generic name for a whole series of binary and multi-phase alloys with other transition elements replacing cobalt. These other elements include copper, chromium, manganese, zirconium and iron. The ratio of transition elements to rare earth may be less than the 8.5 given by the composition. An anisotropic compact is produced and sintered at 1 150 to 1 250°C, and this is followed by a solution treatment at 1 100 to 1 250°C. The compact is then step aged at temperatures from 900°C down to 400°C over a period of at least 20 h. This complicated heat treatment and difficulties with powder production have made the magnets rather difficult to produce with consistent properties. It is however the most stable of the rare earth alloys and has a higher remanence and $(BH)_{max}$ than the $SmCo_5$ alloy.

In both of these alloy systems the properties are due to a very strong magnetocrystalline anisotropy related to the hexagonal crystal structure. The range of sizes which can normally be produced is from about 1 mm cubes up to blocks of 50 × 50 × 25 mm, the short dimension being the preferred direction. These magnets are very brittle and must be handled with care to prevent chipping or cracking. They can only be machined by grinding or slitting, and they cannot be drilled or turned. The magnets may be magnetised prior to assembly without flux loss because they combine a relative recoil permeability close to unity with a high coercivity. However, because of the handling problems with these brittle magnets it is often more convenient for the user to magnetise during assembly.

20.2.4 Neodymium iron boron

This alloy was announced in 1983 and it has generated a great deal of interest because at room temperature it has better properties than any previously known alloy. It is also a cobalt-free alloy and uses neodymium which is more abundant than samarium. These two factors make the raw material costs quite considerably less than those of the samarium cobalt alloys.

There are two main routes for the production of magnets from these alloys. The first is the conventional route similar to that used for $SmCo_5$. The constituents of the alloy are melted together, and the material is then powdered, pressed in a magnetic field and sintered. The basic alloy with this route is the atomic composition $Nd_{15}B_8Fe_{77}$. The second method is to use the melt spinning process where the molten alloy is ejected under pressure onto a water cooled wheel which is rotating at high speed. This cools the alloy very rapidly and an amorphous glass-like tape is produced. This tape is annealed at 650°C, which recrystallises the alloy, and then it is mechanically broken into flake. The basic atomic composition of the alloy for this process is 13.5% Nd 4.75% B 81.75% Fe. There are advantages in each of these methods. To get the best properties the conventional sintering route must be used but the production of the powder is rather slow and expensive. Melt spinning allows the powder to be produced more easily and cheaply. This method is ideal for the production of flake for use in the manufacture of isotropic bonded material, this product being called Magnequench I. The production of anisotropic material from the flake is more difficult and expensive.

The mechanism which provides the high value of the coercivity is crystal anisotropy, in the tetragonal crystals of the alloy. The particle size required for optimum properties is about 3 μm. Powdering can be achieved by ball milling; however the hydrogen decrepitation process has been applied very successfully. This process, which involves the absorption of large volumes of hydrogen by the alloy followed by desorption, produces much less damage to the particles, resulting in better properties.

There are two disadvantages of these neodymium alloys when compared to the rare earth cobalt alloys. They suffer from corrosion and rusting, although this can be prevented by the use of a suitable

coating. They also have a relatively low Curie temperature of 330°C resulting in instability with a rapid decrease in flux with increasing temperature. The stability can be improved by additions of dysprosium, niobium and/or vanadium, making the alloy useful up to 150°C. Similar comments to those for the rare earth cobalt alloys apply here regarding sizes, machining and magnetising but these neodymium alloys are much less brittle.

These alloys are being used in a wide range of rotating machine applications for both a.c. and d.c. generators and motors including stepper motors. They are also used for holding and lifting devices and for motors for compact discs. More specialised applications include body scanners using magnetic resonance imaging (MRI) and particle beam focusing devices.

20.2.5 Bonded materials

Each of the above materials can in the powdered form be mixed with a bond. These bonds can be rubbers, polymers or plastics, and they may be flexible or rigid. The flexible rubber bonded ferrites have found wide application in holding and display services and the best quality anisotropic material is used in small motors. Alnico and rare earth cobalt are also made in bonded forms and they have the advantages of a uniform level of properties and freedom from cracking. Their magnetic properties are not so good as the unbonded materials: at best the BH_{max} is about 50% lower.

For recent research, at the time of writing, in the field of bonded magnetic materials see references 22–24.

20.2.6 Other materials

There are a number of other permanent magnet materials with minor applications. These materials make up less than 1% of the total production. They mainly find applications because they can be mechanically formed into shape, which is not possible for Alnico, ferrites or rare earth alloys. It must be emphasised that these other materials are only available in a limited range of sizes and shapes. Steels which were the original permanent magnets are now almost obsolete because their properties are inferior to those of the materials mentioned above. They do, however, find applications in hysteresis motors where intermediate coercivities are required. Cunife (Cu Ni Fe) and Cr Fe Co are alloys which can be rolled and drawn into wire and the anisotropy is produced by this rolling or drawing. Cr Fe Co produced by rolling can have properties similar to those of anisotropic Alnico. Platinum cobalt (Pt Co), which has a very high coercivity, was much used in the early days of space research. For most applications it has now been replaced by rare earth alloys. However, it is still used for medical and other applications where the resistance to corrosion is required. Other materials which were used in the past are Lodex, which consisted of fine electrolysed particles of iron cobalt, Vicalloy (V Co Fe) and Mn Al C alloys. These materials are now obsolete because of cost, lack of demand and production difficulties.

Non-metallic magnetic materials are of great commercial importance. These are beyond the scope of the current work but are discussed in references 25–27.

20.2.7 Properties, names and applications

The range of properties for each class of material is given in Table 20.6. These ranges include deliberate variations in properties obtained by small changes in composition and heat treatment.

Applications of magnetic materials, especially in electronics, are changing rapidly. See references 28–31 for examples current at the time of writing. Trade names of these materials are not included here, as these are subject to change.

20.3 Magnetically soft materials

Materials or alloys which are classed as magnetically soft, and generally with a relatively high permeability, are used for a wide range of applications including transformer laminations, armature stampings and cores in electrical machinery. They also include materials of considerable importance in the electronic and communication industries. Many special materials of widely different chemical compositions have been developed to meet specialised requirements. These are not confined to metallic alloys and ceramic oxides have become increasingly important.

Table 20.6 CHARACTERISTICS OF PERMANENT-MAGNET MATERIALS

B_t	Remanence, T
$(BH)_{max}$	Energy product, kJ m^{-3}
H_{cB}	Coercivity, kA/m^{-1}
μ_r	Relative recoil permeability

Material	B_r	$(BH)_{max}$	H_{cB}	μ_r
Alnico				
Normal anisotropic	1.1–1.3	36–43	46–60	2.6–4.4
High coercivity	0.8–0.9	32–46	95–150	2.0–2.8
Columnar	1.35	60	60	1.8
Ferrites (ceramics)				
Barium isotropic[a]	0.22	8	130–155	1.2
anisotropic[a]	0.39	28.5	150	1.05
Strontium anisotropic[a]	0.36–0.43	24–34	240–300	1.05
Bonded ferrite				
Isotropic[a]	0.14	3.2	90	1.1
Anisotropic[a]	0.23–0.27	10–14	180	1.05
Rare earth				
$SmCo_5$ sintered[b]	0.9	160	640–700	1.05
$SmCo_5$ bonded[b]	0.5–0.6	56–64	400–460	1.1
Sm_2Co_{17} sintered[c]	1.1	150–240	500–900	1.05
NdFeB[d]	1.0–1.2	200–275	700–835	1.05
Others				
Cunife	0.54	10	40	—
CrFeCo	1.3	40–46	46	3–4
PtCo	0.62	72	370	1.1

Intrinsic coercivity H_{cT}(kA m^{-1}) [a]160–3.10; [b]800–1 500; [c]600–1 300; [d]750–1 900.
The Curie temperature of Alnico is 800–8.50°C, of ferrite 450°C, of SmCo over 700°C and for NdFeB it is 300°C. The Alnicos have a resistivity of about 50×10^{-8} Ωm, for the ferrites it is about 10^4 Ωm and for the rare earths 90–140 $\times 10^{-3}$ Ωm.

Sometimes several conflicting attributes are required in magnetically soft materials for a given application. It is for this reason that a range of materials from iron through to silicon steels, high nickel–iron alloys and ferrites have emerged. Usually, the most important requirement is that the hysteresis loss due to cyclic magnetisation should be as low as possible. The eddy current loss, which is due to the induction of electric currents in the material by the changing magnetic flux, should also be small. Generally the magnetic permeability should be as high as possible, particularly the maximum permeability, and in certain special cases the initial permeability also. Sometimes the requirement is for a high permeability at low field strengths. In general, the saturation intensity should be as high as other conditions will permit. For magnetic temperature compensation there should be a linear relation, over a known range of temperature, between permeability and temperature.

Relevant data concerning a wide selection of materials are given in Tables 20.7–20.11 and are discussed in the corresponding sections.

20.3.1 Silicon–iron alloys

For the magnetic circuits of the larger pieces of electrical equipment normally operated at power frequencies of 50 Hz, such as power transformers, alternators and electric motors of all sizes, a magnetic material is essential which will have a low hysteresis, high resistivity and a high value of magnetic saturation. Its production in the form of thin sheet or strip must also be possible, and in view of the big tonnages essential for such purposes it must not be expensive. Silicon–iron alloys are the only ones which meet all these conflicting requirements.

Over the years, the quality of silicon steels has been continuously improved and more has been learned about the effects of silicon on the magnetic properties. These effects are briefly as follows:

1. Silicon precipitates the major portion of the oxygen dissolved in the steel, which reduces hysteresis loss and magnetic ageing.
2. Silicon increases the electrical resistivity and thus reduces the eddy current loss, which is proportional to the resistivity (*see* Figure 20.3).

LIVE GRAPH
Click here to view

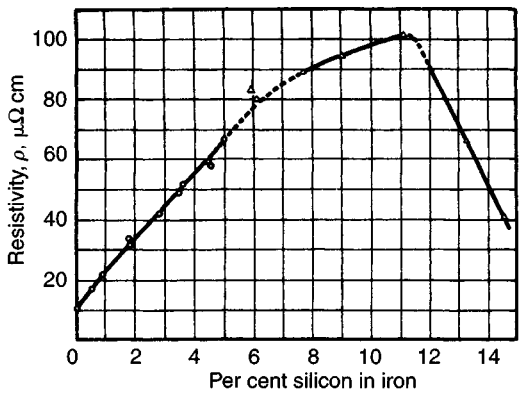


Figure 20.3 *Electrical resistivities of iron-rich iron–silicon alloys*

LIVE GRAPH
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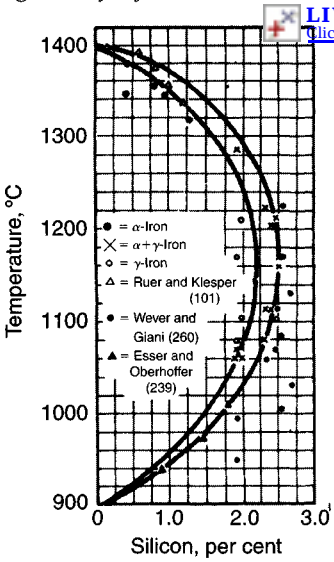


Figure 20.4 *The gamma phase region*

3. With over 2.5% silicon the alloys are outside the gamma loop on the equilibrium diagram, i.e. the alpha–gamma change point is suppressed (*see* Figure 20.4). Alloys of more than 2.5% silicon can therefore be annealed at high temperatures without recrystallisation occurring on cooling. As a result it is easier to produce large grain size in the finished sheet, with a corresponding reduction in hysteresis loss.
4. Silicon precipitates the carbon as graphite if it exceeds about 0.05%. Carbon present as graphite has no effect on hysteresis loss.

It is possible to orient the grains in silicon steels by cold rolling and heat treatments. Improvements in this grain-oriented sheet have been made as follows:

1. Limitation of the silicon content to about 3%. Above this value it is difficult to produce good quality sheet because of mechanical difficulties.
2. Reduction in the sheet thickness. The minimum thickness is 0.23 mm, which is limited by grain growth considerations.
3. Reductions of impurities to the minimum economic value. Impurities which increase hysteresis are those which occupy interstitial positions in the crystal lattice, e.g. carbon, oxygen, nitrogen and sulphur.
4. Improvements in the control of annealing equipment and protective atmospheres during final annealing.

The grain-oriented steel strip is used for transformers. The grain orientation is parallel to the rolling direction and in the transformer the flux is arranged so that it is parallel to this direction. This gives the advantage that the permeability is better and the hysteresis is lower with the direction of magnetisation parallel to the cube edge of the crystal lattice (*see* Figure 20.5).

During the annealing process a glass film is produced on the steel surface and this film gives a longitudinal tension in the sheet and provides insulation. This tension improves the magnetic properties of the steel, including a reduction in the magnetostriction which reduces the acoustic noise of the transformer. A phosphate coating is applied to complete the tensioning and insulation. The grain size of these sheets is comparatively large and the resulting domain boundaries are quite widely spaced. Artificial grain boundaries can be produced by laying down lines of ablated spots on the steel surface to give a stress and atomic disruption pattern which pins domain walls and leads to a smaller wall spacing. Various methods have been used to produce this ablating including spark and laser techniques.

In addition to the grain oriented material there is a need in rotating machinary applications for sheets which are comparatively isotropic and less expensive. The silicon content is generally lower as

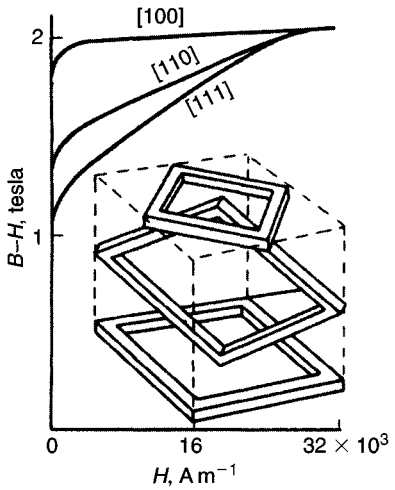


Figure 20.5 *Magnetisation curves for 3.85% silicon-iron single crystals at high field strengths in three directions*

the minimum core loss is less important than the high-saturation magnetisation needed for maximum torque.

Manufacturers offer a series of alloys whose properties are given in Table 20.7. These alloys conform to the British Standards and their international equivalents given in ref. 38. Methods of test are also given in these standards. In the past the Epstein square has been used by there is a rapidly growing trend towards the use of single-sheet testers which greatly reduce the labour involved.

20.3.2 Ferrites[34–36] and garnets[36]

Ferrites are iron oxide compounds usually containing at least one other metallic ion. The magnetic iron oxides Fe_3O_4 and Fe_3O_3, both of which are used in magnetic recording, are also included with the ferrites. The most important of the many ferrite materials now known are sintered ceramics of high resistivity, together with saturation magnetisations up to 0.5 T at room temperature. One structural class is the barium hexaferrites with high coercivities used as permanent magnets. The other two structural classes are the spinels and the garnets.

The spinel-structured ferrites are mixed oxides with the general composition $MO \cdot Fe_2O_3$ where M may be Fe, Ni, Mn, Zn, Co, Cu, Li or Al or mixtures of these. They are prepared from the constituent oxides which have been milled to a fine powder, preferably of the same particle size and intimately mixed. This mixture is calcined at about 1 000°C followed by crushing and milling and pressing the powder in a die or extrusion to the required shape. The resulting compact is a black brittle ceramic and any subsequent machining must be by grinding. They may be prepared with high permeability, and by virtue of the high resistivity, which renders eddy current levels negligible, can be used at frequencies up to 10 or 20 MHz as the solid core of inductors or transformers. The losses which do occur are a combination of hysteresis, eddy current, and residual losses which may be separately controlled by composition and processing conditions with due regard to the permeability required and frequency of operation.

The saturation flux density of ferrites is low, making them unsuitable for use in power or high flux transformers. Their use is therefore almost entirely in the electronic and telecommunications industry where they are now largely replacing laminated alloy and powder cores.

Typical properties of the commonly employed manganese–zinc and nickel–zinc ferrites are given in Table 20.8.

For higher frequencies of 100 mHz and above garnets are used. They have resistivities in excess of 10^8 Ωm compared with the ferrites, restivities of about 10^3 Ωm at best. This means that eddy current losses which are dependent on resistivity are minimised. The basic Yttrium Iron Garnet (YIG) composition is $3Y_2O_3 \cdot 5Fe_2O_3$. This is modified to obtain improved properties such as very low loss and greater temperature stability by the addition of other elements, including aluminium

Table 20.7 TYPICAL PROPERTIES OF SILICON STEELS[32]

Grade identity	*Thickness* mm	*Typical specific total loss at* $B = 1.5$T; 50 Hz (*or* 1.7T; 50 Hz *where stated*) (W kg)	*Typical specific apparent power at* $B = 1.5$T; 50 Hz (*or* 1.7T; 50 Hz *where stated*) (VA kg^{-1})	*Nominal silicon content* %	*Resistivity* $\Omega m \times 10^8$	*Stacking factor* %	$\hat{B}$ *at* $H = 5\,000$ A m^{-1} (*or* 1 000 A m^{-1} *where stated*) T	*Typical applications*
Grain oriented: magnetic properties measured in direction of rolling only								
Unisil								
27 MOH	0.27	1.03 (1.7; 50)	1.4 (1.7; 50)	2.9	45	96	1.94 (1 000)	High-efficiency power transformers
30 M2H	0.30	1.12 (1.7; 50)	1.55 (1.7; 50)	2.9	45	96.5	1.92 (1 000)	
Unisil								
23M3	0.23	0.75	1.00	3.1	48	96	1.85 (1 000)	
27M4	0.27	0.84	1.16	3.1	48	96	1.85 (1 000)	
30M5	0.30	0.89	1.32	3.1	48	96.5	1.85 (1 000)	
35M6	0.35	1.00	1.39	3.1	48	97	1.84 (1 000)	
Non-oriented: magnetic properties measured on a sample comprising equal number of strips taken at 0° and at 90° to the direction of rolling								
Transil 300-35-A5	0.35	2.95	30	2.9	48	98	1.65	Large rotating machines
Losil 400-50-A5	0.50	3.60	19	2.4	44	98	1.69	Small transformers/chokes
Losil 800-65-A5	0.65	6.50	14	1.3	29	98	1.75	Motors and FHP motors
Non-oriented: grades supplied in the 'semi-processed' condition and which require a decarburising anneal after cutting/punching to attain full magnetic properties								
Newcor 800-65-D5	0.65	7.00	10	Nil	12	97	1.77	Motors and FHP motors
Newcor 1000-65-D5	0.65	8.80	11.5	Nil	12	97	1.74	
Tensile grades								
Tensiloy 250	1.60	—	—	Nil	—	—	1.60	Pole pieces large rotating machines

Table 20.8 TYPICAL PROPERTIES OF SOME MAGNETICALLY SOFT FERRITES*[34–36]

	Conditions		*Ferrite grade*					
Parameter	f kHz	$\hat{B}$ mT	(i)	(ii)	(iii)	(iv)	(v)	*Units*
Approximate composition:								
$MnO/ZnO/Fe_2O_3$			27/20/53	25/22/53	30/20/49	34/14/52	—	mol %
$NiO/ZnO/Fe_2O_3$			—	—	—	—	32/18/50	mol %
Initial permeability, μ_i	<10	<0.1	1 200–2 500	3 800–10 000	350–500	1 000–3 000	70–150	
Temperature factor, $\Delta\mu_i/\mu^2\Delta\theta$	<10	<0.1	0.5–1.5	—	20	—	0–8	$°C^{-1} \times 10^{-6}$
(5 to 55°C)	30		0.8–2.0	5–10	—	—	—	
	100		1.5–3.0	10–50	30	—	—	
Residual loss factor (tan δ_r)/μ_i		<0.1						10^{-6}
	10^3		—	—	75	—	20–40	
	10×10^3		—	—	—	—	60–100	
Hysteresis coefficient, η	10	1–3	0.5–1.0	0.1–1.0	—	—	2–10	$mT^{-1} \times 10^{-6}$
Saturation flux density, B_{sat} (at 1 kA m^{-1})			0.35–0.45	0.35–0.45	0.23–0.27	0.4–0.5	0.3–0.4	T
Curie point θ_c	<10	<0.1	130–250	90–220	130–160	180–280	250–400	°C
Resistivity	d.c.		1–5	0.05–0.5	1–5	1–5	$>10^3$	Ωm
Applications			Quality inductors	Wide-band or pulse transformers	Deflection yokes	High-power transformers	Quality inductors 2–12 MHz	

* At 25°C unless otherwise stated.

and gadolinium. The garnets are prepared by heating the mixture of oxides under pressure at over 1 300°C for up to 10 h. This material is used in microwave circuits in filters, isolators, circulators and mixtures. Some of their properties are given in Table 20.9.

Table 20.9 GARNET MATERIAL

	J_s T	T_c °C	ΔH (kA m^{-1})* Maximum
Y Al Fe	0.3–1.76	110–280	45–60
Y Gd Fe	0.67–1.22	280	110–250
Y Gd Al Fe	0.6–1.1	190–250	70–120

* This is the gyromagnetic resonant linewidth ΔH (−3 dB).
The values of other parameters for these garnets are:
dielectric constant ε' 13.8–15.2,
$10^4 \tan\delta$ 2.5–3.

20.3.3 Nickel–iron alloys

The high permeability of nickel–iron alloys makes them very useful materials for the electronic communications and electrical power engineering industries. This high permeability at low fields (despite the low saturation magnetisation) gives scope for minimising core sizes in a wide range of applications.[32,33]

There are three main groups of high-permeability nickel–iron alloys. The 80% nickel alloys variously called Mumetal and Permalloy have the highest permeability due to the almost zero crystalline anisotropy and magnetostriction at this composition. The commercial alloys have additions of chromium, copper and molybdenum to increase the resistivity and improve the magnetic properties. This material is used in specialised transformers, for magnetic screens and magnetic recording heads.

The 50% nickel alloys have the highest saturation magnetisation of 1.6 T in this range of alloys. They have a higher permeability and better corrosion resistance than silicon–iron alloys but they are more expensive. Another advantage is they have a high incremental permeability over a wide range of polarising d.c. fields. A wide range of properties can be produced by various processing techniques. Severe cold reduction produces a cube texture and a square hysteresis loop in annealed strip. The properties can be tailored by annealing the material below the Curie temperature in a magnetic field. Applications of this material include chokes, relays and small motors.

The 36% nickel alloys have higher resistivity and are less costly than the 50% alloy but the saturation magnetisation and permeability are lower. This alloy is of the Invar type and has a very low coefficient of expansion of 1×10^{-6}°C^{-1}. The main applications are relays, inductors and high-frequency transformers where the higher resistivity limits the eddy currents. The properties of the various nickel–iron alloys are given in Table 20.10.

Table 20.10 PROPERTIES OF NICKEL–IRON ALLOYS[32]

Alloy	36% Ni	45% Ni	50% Ni	54% Ni	65% Ni	77% Ni
Initial permeability $\times 10^{-3}$	3	6	0.5–11	65	—	60–200
Saturation flux density (T)	1.2	1.6	1.54–1.6	1.5	1.3	0.8
Coercivity force (Am^{-1})	10	10	3–10	2.5	2–10	0.5–1
Remanence (T)	0.5	1.0	1.1–1.5	0.7	1.25–0.2	0.5
Resistivity ($\mu\Omega$ m)	0.8	0.45	0.4	0.45	0.6	0.6
Density (kg m^{-3})	8 100	8 300	8 300	8 300	8 500	8 800
Curie temp. (°C)	280	530	530	550	520	350
Expansion coefficient (°C^{-1} $\times 10^6$)	1	8	10	11	12	13

20.3.4 Amorphous alloy material

The rapidly solidified alloy materials called metallic glasses are now an established group of soft magnetic materials. These materials are produced by rapid solidification of the alloy at cooling rates around a million degrees centigrade per second. The alloys solidify with a glass-like atomic structure which is a non-crystalline frozen liquid. The rapid solidification is achieved by causing the molten alloy to flow through an orifice on to a rapidly rotating water cooled drum. This can produce sheets of the alloy as little as 10 μm thick and up to a metre or more wide.

There are two main groups of amorphous alloys. The iron-rich magnetic alloys have the highest saturation magnetisation among the amorphous alloys and are based on inexpensive raw materials. The other group are the cobalt-based alloys which have very low or zero magnetostriction which leads to the highest permeability and the lowest core loss. All of these alloys have high electrical resistivity, higher than that of conventional crystalline electrical steels. Because of this, eddy current losses are minimised both at 50 Hz and at higher frequencies. They also have other advantages including flexibility without loss of hardness, high tensile strength and better corrosion resistance than similar crystalline alloys. Properties of typical amorphous alloys are given in Table 20.11.

Table 20.11 AMORPHOUS ALLOYS[2]

Amorphous material	J_s T	*Curie temperature* °C	*Crystal-lisation* °C	*Electrical resistivity* μΩ m	μ_{max}, d.c. annealed	*Coercivity* A m^{-1}	*Saturation magneto-striction* $\lambda_s \times 10^6$
Iron-based	1.28–1.80	310–415	430–550	1.23–1.37	35–600 × 10^3	1–1.6	19–35
Nickel–iron-based	0.88	350	410	1.38	8 × 10^5	0.4	12
Cobalt-based	0.55–0.70	200–365	520–550	1.36–1.42	6–10 × 10^5	0.15	<1

20.4 High-saturation and constant-permeability alloys

The addition of cobalt to iron increases the saturation intensity. A range of alloys with 24–50% cobalt has saturation up to 10% greater than that of iron. Alloys with around 25% cobalt are used for applications such as magnet pole tips where a combination of good ductility, ease of machining and high-saturation magnetisation are required. The permeability and loss characteristics are however inferior to the 50% cobalt alloy. The 49/49/2 Fe/Co/V alloy called Permendur, to which the vanadium has been added to improve the ductility and also substantially increase the resistivity, is used extensively for stators in lightweight generators.

Recent developments, which have improved its mechanical properties, have extended its use to rotor laminations. Other applications include small-volume transformers, relays, diaphragms, loudspeakers and magnet pole tips. The high cost of cobalt restricts the use of these alloys, but they are widely used in aircraft where weight is at a premium. The properties of the alloys are given in Table 20.12.

Table 20.12 IRON–COBALT ALLOYS[32]

Alloy	*Saturation flux density* T	*Initial permeability*	*Maximum permeability*	*Coercive force* A m^{-1}	*Remanence* T	*Resistivity* μΩ m	*Curie temperature* °C
24% Co	2.35	250	2 000	130	1.5	0.2	925
49/49/2 Co/Fe/V	235	800	7 000	950	1.6	0.4	975

Constant-permeability materials are required for applications where a constant inductance is essential for electrical filters or in loading coils for telephone lines. The original materials were

in the ternary Fe/Ni/Co series but more recently they have been replaced by the 40–50% nickel Isoperms. They have an addition of 5–15% copper and by cold rolling and heat treatment a preferred orientation is produced. The permeability of 50–60 is constant over a range of fields up to 8 kA m^{-1}.

20.5 Magnetic powder core materials

The thickness of laminations of ferromagnetic alloy cores must be reduced as the operating frequency is increased above audio range if the increasing loss and reducing permeability caused by eddy currents are to be avoided. However, reduction of thickness of laminations causes increased cost. This may be avoided for many applications up to 200 kHz by the use of magnetic powder cores.[3]

The Permalloys are used in the form of compressed insulated powder cores, the effect of subdivision being to reduce eddy current losses to a minimum. The permeability is also reduced but is made extremely constant, becoming within a small percentage independent of magnetising force and unaffected by magnetic shock. The properties of such materials are in many ways similar to the constant permeability alloys described in the preceding section. *Sendust*, a brittle alloy of aluminium, silicon and iron, has been similarly used in Japan and Germany. Carbonyl iron powder is used for powder cores mainly at radio frequencies, specially low eddy current losses being obtained on account of the characteristic fine particle size. Some details of these materials are given in Table 20.13.

Table 20.13 MAGNETIC POWDER CORE MATERIALS

Material	*Composition*	*Effective initial relative permeability* μ_e	*Typical power loss* kw/m^3 (*at* 50 kHz, 100 mT, 25°C)	*Typical applications*
Molybdenum Permalloy	80% Ni–Fe +Mo	125 60	230 500	High Q fixed inductors 0.1 to 200 kHz
Radiometal (high flux)	50% Ni–Fe	125	600	Switch mode power supply chokes and transformers to 250 kHz
		69	650	Mains chokes and filters for suppression of radio frequency interference
Carbonyl iron powder (high frequency)	99.9% Fe	5–25	—	High Q fixed inductors 1–300 MHz. Wide band signal transformers up to 10 GHz. Chokes for RF suppression
Electrolytic iron powder (low frequency)	99% Fe	45–90	3 300	Switch mode power supply chokes to 50 kHz. Main chokes for RF suppression
			2 400	Dimmer chokes 50 Hz DC electromagnets
Sendust	9.5% Si, 5.5% Al, balance Fe	60	2 000	Chokes and inductors

20.6 Magnetic temperature-compensating materials

Alloys having a marked variation of permeability with temperature are used to compensate for errors in electrical instruments caused by changes in the ambient temperature.

The magnetic permeability of iron and iron alloys decreases with temperature (except for low values of the field), and this causes the readings of electrical meters which depend upon the fluxes maintained by a constant voltage or current or by a permanent magnet, to decrease with the ambient temperature. To compensate for such errors it is customary to shunt a certain proportion of the magnetic flux passing through the magnetic circuit of the meter by means of an alloy having an unusually high negative temperature coefficient between 0 and 100°C, so that, as the ambient temperature increases, the proportion of the shunted flux decreases, causing more flux to pass through the working air gap of the instrument than otherwise would be the case. By correctly proportioning

the shunt component it is possible to render the readings almost completely independent of ambient temperature changes.

Compensation may also be provided for the changes of resistivity of eddy current devices. These alloys have a major application in the compensation of changes in the resistivity with temperature of the copper drag cup in speedometers.

The alloys used are in the nickel–iron series, about 30% nickel, sometimes with additions of chromium, manganese and/or silicon. Details of the characteristics of the alloy are given in Table 20.14.

Table 20.14 THERMOMAGNETIC TEMPERATURE COMPENSATING MATERIALS

	Magnetic flux density at $H = 8 \times 10^3\ \mathrm{A\,m^{-1}}$ T								
	Temperature °C								
Material	−40	−20	0	20	40	60	80	100	120
Telecon 2799	0.53	0.43	0.33	0.22	0.13	0.05	—	—	—
Telecon 2800	—	—	0.76	0.67	0.59	0.50	0.40	0.29	0.19
Thermoflux 90/100 G	—	—	0.60	0.47	0.33	0.20	0.07	—	—
Thermoflux 65/100 G	—	—	0.43	0.31	0.17	—	—	—	—
Thermoflux 35/100 G	—	—	0.19	0.07	—	—	—	—	—
Carpenter 30	0.60	0.51	0.40	0.28	0.14	0.05	—	—	—
Carpenter 32	1.02	0.97	0.90	0.83	0.75	0.67	0.59	—	—

20.7 Non-magnetic steels and cast irons

For certain constructional purposes in electrical engineering where magnetic shunting effects have to be avoided there arises a requirement for almost non-magnetic materials possessing certain minimum mechanical properties.

The materials available comprise non-magnetic cast irons, austenitic steels, and certain other sensibly non-magnetic alloys; the choice of materials for any particular purpose is dependent upon the mechanical properties required; e.g. certain intricate stationary parts of electrical machinery would be suitably made from non-magnetic cast iron, while armature straps for high-speed rotors would be made in a high-tensile austenitic alloy steel in order to withstand the dynamic stresses encountered in practice. A high resistivity is sometimes beneficial as well as the possession of virtually non-magnetic properties.

There are two well-known non-magnetic cast irons, one called *Nomag*, a manganese-silicon-nickel-iron with a permeability of 1.04, and a resistivity of 1.40 microhm *m*, and the other *Ni-Resist*, containing lower proportions of manganese and silicon, but a higher proportion of nickel, also a little chromium and copper. These irons with flake graphite are brittle, and can only be used for parts of a simple nature, such as supports which are not subjected to severe mechanical stress. These non-magnetic irons are also made in the spheroidal graphite form e.g. *Nodumag* and *S. G. Ni-Resist*, having the characteristically improved mechanical properties of this type of cast iron.

As examples of austenitic steels. Hadfields 13% manganese steel and the chromium–nickel steels have found considerable use. The stainless steels of the chromium–nickel type are normally non-magnetic but if they are subjected to cold working appreciable magnetic properties may appear. This disadvantage can be overcome by using steels with higher alloy content than the well-known 18/8 type. Besides these materials other non-magnetic austenitic steels have been specially developed to give improved mechanical properties.

Although the originators of some of these steels and cast irons are no longer in business, it is possible to get steelmakers or foundries to produce them to order. Table 20.15 gives details of the non-magnetic cast irons and a number of non-magnetic cast steels.[37]

Table 20.15 NON-MAGNETIC STEELS AND CAST IRONS

Material	*Nominal composition*	*Relative permeability*	*Resistivity* $\mu\Omega$ m	*Condition*	*Mechanical properties*				
					Hardness HB	*Proof stress* MPa	*Max stress* MPa	*Elongation* L=4$\sqrt{A}$ %	*Reduction in area* %
Wrought steels									
18/8 Stainless	8% Ni, 18% Cr and various minor constituents	1.06 or greater	0.72	Austenised	170	278	618	45	50
18/8 Stainless	8% Ni, 18% Cr and various minor constituents	1.4	—	20% cold reduction	280	—	927	25	—
18/8 Stainless	8% Ni, 18% Cr and various minor constituents	about 10.0	—	50% cold reduction	330	—	1 313	10	—
18/12 Stainless	12% Ni, 18% Cr etc.	1.03	0.80	Austenised	160	232	618	60	65
18/12 Stainless	12% Ni, 18% Cr etc.	1.1	—	50% cold reduction	—	—	1 313	—	—
18/12 Stainless	12% Ni, 18% Cr etc.	1.7	—	90% cold reduction	—	—	—	—	—
25/12 Stainless	12% Ni, 25% Cr etc.	1.01	0.90	Austenised	200	309	695	30	35
25/12 Stainless	12% Ni, 25% Cr etc.	1.03	—	90% cold reduction	—	—	—	—	—
Foxalloy 101	1% Si, 16–20% Mn, 12–16% Cr, 0.2–1% V	1.01 max	0.75	Austenitic	275	750	960	35	52

(*continued*)

Table 20.15 NON-MAGNETIC STEELS AND CAST IRONS—*continued*

Material	*Nominal composition*	*Relative permeability*	*Resistivity* $\mu\Omega$ m	*Condition*	*Mechanical properties*				
					Hardness HB	*Proof stress* MPa	*Max stress* MPa	*Elongation* L=4$\sqrt{A}$ %	*Reduction in area* %
Cast steels									
Hadfields Mn	13% Mn, 1% Si	1.03	0.71	Water quenched 1 000°C	—	—	—	—	—
Clyde alloy non-magnetic	8% Ni, 8% Mn, 8% Cr	1.003	0.76	Water quenched 900°C	—	—	—	—	—
Clyde alloy non-magnetic	8% Ni, 4% Mn, 8% Cr	1.003	—	Water quenched 900°C	—	571	927	45	45
Firth Vickers NMC	10% Ni, 5% Mn, 4% Cr	1.008/1.03	—	—	—	618	927	45	45
Black alloy 525 (tool bit material)	48% Co, 35% Cr, 14% W carbide, 9% Nb/Ta	1.002/1.01	—	—	620	—	450	—	—
Cast irons									
Nomag	7% Mn, 11% Ni, 2.5% Si, 2.7% C	1.03/1.05	1.40	Stress relieved	120/150	—	~ 185	0	—
Nodumag	7% Mn, 11% Ni, 2.5% Si, 2.7% C	1.03	1.10	Stress relieved	120/150	309	432	15	—
Ni-Resist Type 2	Ni 20% Si 2.5%	1.03	1.70	Stress relieved	150	—	185	0	—
S. G. Ni-Resist Type D-2	Mn 1% Cr 2% C 2%	1.03	1.00	Stress relieved	140	232	415	15	—

Appendix

Units and definitions

Magnetic poles

The concept of magnetic poles is used to explain the behaviour of magnetised bodies acting on one another and in the earth's magnetic field. Like magnetic poles repel each other and unlike poles attract.

Magnetic dipole

A magnetic dipole is an elementary pair of unlike poles and can be represented by an infinitesimal current loop.

Magnetic moment m

For a magnetic dipole the magnetic moment is the product of the current and the loop area. For a magnetised body the magnetic moment is the sum of all the magnetic dipole moments and is equal to ml where l is the distance between the poles.

Magnetic field strength H

The magnetic field strength H at a point is an axial vector quantity associated with the magnetic flux density in a magnetic field. The unit of magnetic field is the ampere per metre ($\mathrm{A\,m^{-1}}$), which is the (axial) field inside an infinitely long solenoid of one turn per metre carrying a current of one ampere.

Magnetisation M

When a magnetic field is applied to a magnetic material the material becomes magnetised. The magnitude of this magnetisation M is equal to the total magnetic moment divided by the volume. The units of M are ampere per metre ($\mathrm{A\,m^{-1}}$).

Polarisation J

This is the name given to the product $\mu_0 M$, and it has the units of tesla (T).

Saturation magnetisation M_s *and polarisation* J_s

These are the maximum possible values of the magnetisation and polarisation for a given material at a given temperature. The saturation polarisation J_s is sometimes called B_s.

Magnetisation per unit mass σ

The value of σ is usually the saturation value and is equal to M/ρ where ρ is the density. The units are joule/(tesla kg).

Magnetic flux density B (or magnetic induction)

The magnetic flux density B at a point is a vector quantity arising from the magnetic field and the magnetisation of the medium. It is given by:

$$B = \mu_0 H + \mu_0 M$$

or

$$B = \mu_0 H + J$$

where $\mu_0$is the magnetic constant. The units of B are tesla (T) or weber/(metre)^2 ($\mathrm{Wb\,m^{-2}}$).

Magnetic constant μ_0

This constant μ_0 is equal to $4\pi \times 10^{-7}$ Henries $\mathrm{m^{-1}}$. For practical purposes this constant when multiplied by the field strength equals the flux density in vacuo. (It was in the past called the permeability of free space.)

Magnetic flux ϕ

The magnetic flux ϕ is the sum over an area A of the magnetic flux density and $\phi = BA$. The unit of flux is the weber (Wb).

Magnetic susceptibility κ

The magnetic susceptibility κ of a material is the ratio of the magnetisation M to the magnetic field H producing it. This quantity κ is the magnetic susceptibility per unit volume and is dimensionless.

Mass susceptibility χ

The magnetic susceptibility per unit mass or mass susceptibility χ is given by κ/ρ, where ρ is the density of the material. It has the units of $m^3\ kg^{-1}$.

Magnetic permeability μ

The permeability μ of a medium is the ratio of the magnetic flux B to the magnetic field H producing it. For practical purposes the value of μ is divided by the magnetic constant μ_0 to give the relative magnetic permeability μ_r which is dimensionless. For strongly magnetic materials the value of μ_r varies with the value of the magnetic field. The values of μ_i, the initial permeability, and μ_{max}, the maximum permeability, are widely quoted parameters for magnetic materials.

Curie point T_c

The Curie point, or temperature, (T_c) is the temperature below which a material is ferromagnetic or ferrimagnetic and above which it is paramagnetic.

Néel point N_c

The Neel point or temperature is the temperature below which a material is antiferromagnetic and above which it is paramagnetic.

Magnetic hysteresis

This is a phenomenon by which the magnetisation of a material depends not only on the magnetising field but also on the previous magnetic history. The magnetic flux B always lags behind magnetic field H in a material showing hysteresis. The complete B–H curve, or loop, produced by increasing H from zero to a maximum value, reducing to zero and then repeating with H in the negative sense is termed the hysteresis loop, and the dissipation of energy associated with it, the hysteresis loss. The hysteresis loss per unit volume is given by (area of the loop in $B \times H$) joules.

Remanence B_r

This is the value of the magnetic flux density remaining in a material in a closed magnetic circuit following magnetisation to saturation and subsequent removal of the magnetic field.

Coercivity H_c

This is the magnetic field strength when a material is brought from saturation to the point where the flux density B or magnetisation M is reduced to zero. In a graphical representation it is the value corresponding to the intersection of the B curve or the M with the H axis. Two cases must be distinguished. When the value of B is reduced to zero the coercivity is called H_{cB} and when the value of M (or J) is reduced to zero the value is called H_{cM} (or H_{cJ}). The quantity H_{cJ} is sometimes called H_{ci}.

Energy Product and $(BH)_{max}$

The energy product is the product of the flux density and the field strength at any point of a demagnetisation curve for a permanent magnet. The maximum value of the product is called $(BH)_{max}$. It is directly related to the stored energy per unit volume of the material. The units of $(BH)_{max}$ are joules/(metre)^3.

Magnetostriction

Elastic deformation which accompanies a change in the magnetisation of a material. This results in changes of length or volume.

Magnetic units and conversion factors

Authors and publishers of current literature on magnetism and magnetic materials have been very slow in adopting SI (systeme international) units in place of the centimetre gram second (c.g.s.) electromagnetic units previously used. Conversion of magnetic units between the two systems are therefore frequently required, as well as care in using equations which are different for the two systems. The relation between the relevant c.g.s. electromagnetic units and the SI (rationalised m.k.s.) units is given in Table 20.16.

Table 20.16 RELATION BETWEEN SI UNITS AND .C.G.S. UNITS OF MAGNETIC QUANTITIES

Quantity	*c.g.s. unit*	*SI unit*	*Conversion factor**
Magnetic field H	Oersted (Oe)	amperes per metre ($A\,m^{-1}$)	$\times 10^3/4\pi$
Magnetic flux ϕ	Maxwell (or line)	Weber (Wb)	$\times 10^{-8}$
Flux density B	Gauss (Gs)	Tesla (T)	$\times 10^{-4}$
Magnetomotive force	Oersted cm or Gilbert	amperes	$\times 10/4\pi$
Polarisation	I *or* J e.m.u.	J Tesla (T)	$\times 4\pi \cdot 10^{-4}$
Magnetisation	I *or* J e.m.u.	M amperes per metre	$\times 10^3$
Maximum energy product $(BH)_{max}$	M Gs Oe	$J\,m^{-3}$ or $T\,A\,m^{-1}$	$\times 10^5/4\pi$
Magnetic moment m.	e.m.u.	Weber metre (Wb m)	$\times 4\pi 10^{-10}$
Magnetic constant μ_0	1	Henry m^{-1} ($H\,m^{-1}$)	$\times 4\pi 10^{-7}$
Resistivity ρ	ohm · cm	ohm · m	$\times 10^{-2}$

* The number by which the quantity in c.g.s. units must be multiplied to obtain the quantity in SI units.

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